RETRO COMPUTER FESTIVAL 2026

LUNA STATION

A home-made retro console on a tiny FPGA — featuring Lunar Lander



Luna Station is a hobby project celebrating **60 years since Luna 9** made the first soft landing on the Moon in 1966. Built for the **Retro Computer Festival in Cambridge**, it reimagines Lunar Lander on a computer designed from scratch.

HOW TO PLAY

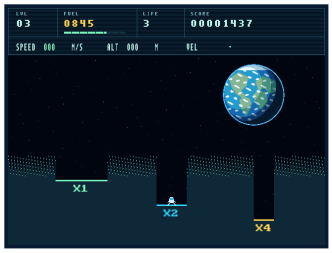
Rotate the lander, fire the engine and touch down on a flat pad before the fuel runs out. Land too fast or at an angle and it's a crash. Narrow pads score **×2** or **×4** — if you dare.

WHAT'S INSIDE?

There is no ready-made graphics chip and no operating system here. An **FPGA** is a chip full of blank logic that can be "wired" by software. We described a complete little computer in Verilog: a processor, memory, a video card and an SD-card reader — and the chip became that computer.

> The game runs with just **8 KB of RAM** — the same as an Atari 400 from 1979 — and does all its physics with integer "fixed-point" maths, just like the classics.





Real screenshots: main menu and a flight over the landing pads (640×480 HDMI).

TECH SPECS

BOARD	Sipeed Tang Nano 9K · Gowin GW1NR-9 FPGA, 8640 logic cells
CPU	PicoRV32 — an open-source **RISC-V** processor, 32-bit, 27 MHz
VIDEO	HDMI 640×480 @ 60 Hz generated in pure logic; lander sprite drawn in 32 rotations
STORAGE	Games load from an SD card (FAT) and can be updated over USB
SOUND	ESP32 synthesizer, six-voice music, I²S amplifier MAX98357A
TOOLS	Yosys, nextpnr, GCC; a 3D-printable Luna Station case design

A TRIBUTE

Leave the console idle and it remembers **Luna 9** (1966), the first soft landing on the Moon, and **Apollo 11** (1969), the first people to walk there.

SCAN FOR THE SOURCE CODE, SCHEMATICS & 3D FILES

github.com/melrouts/RetroConsole-LunaStation

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A D ROTATE
B MENU

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